

A High-Throughput and Memory-Efficiency 2-D DCT Architecture Based on CORDIC Rotation

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Abstract

2-D Discrete Cosine Transform (DCT) applies on image data compression and saves more memories. In this paper, we use fast DCT algorithm, and propose a parallel-pipelined architecture to implement a 8×8 DCT/IDCT processor. This architecture involves two 8-point DCT processors, dual-bank of SRAM (128 words) and the coefficient ROM, three multiplexers, timing controller and 7-bit counter. The kernel arithmetic unit (AU) of 1-D DCT processor is designed by the CORDIC arithmetic. This architecture saves hardware and power consumption, and achieves high performance.

1. Introduction

For the application of image data compression, a transform encoding technique is more suitable than that of a linear prediction coding (LPC). It is clear that the Walsh-Hadamard transform is the simplest one for implementation. The kernel matrix of the Walsh-Hadamard transform involves only additions and subtractions [1]. However, the Walsh-Hadamard transform is definitely not the best transform encoder due to the absence of symmetry property of the transform kernel. Currently, the Discrete Cosine Transform (DCT) becomes the best transform encoder because of that the transform kernel of the DCT involves both the periodicity and symmetry properties, and therefore the DCT can perform very close to the theoretically optimal Karhunen-Loeve transform [2]. Conventionally, a $2N$ -point DCT or the double size Fast Fourier Transform (FFT) computation algorithm was employed to the implementation of DCT involving complex arithmetic [1], where the processor units as well as the computation time were $2N$ and $3N$, respectively required for the implementation of DFT algorithm, and were $2\log 2N$ and $O(\log 2N + 1)$, respectively required for the implementation of FFT algorithm. Some fast computation algorithms were proposed and discussed [3]-[11], and the VLSI chips implementations of some DCT concurrent architectures were accomplished for real-time applications [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22].

In this paper, a CORDIC-based fast computation approach for the DCT implementation will be investigated. In section 2, the CORDIC algorithm will be described. In section 3, 2-D CORDIC-based fast DCT algorithm is presented. In section 4, the implementation of low-power parallel-pipelined 2-D DCT architecture is presented. Finally, conclusions will be given.

2. CORDIC Algorithm

CORDIC (COordinate Rotation DIgital Computer) is an algorithm for performing a sequence of micro-rotation using a coordinate rotation [23] [24]. In a circular coordinate system, the rotation by an angle θ is implemented as an iterative process,

consisting of micro-rotations during which the initial vector is rotated by pre-determined step angles $\alpha_i = \arctan(2^{-i})$. The basic CORDIC iteration in a circular coordinate system is described as:

$$x_{i+1} = x_i - \sigma_i 2^{-i} y_i \quad (1)$$

$$y_{i+1} = y_i + \sigma_i 2^{-i} x_i \quad (2)$$

$$z_{i+1} = z_i - \sigma_i \alpha_i \quad (3)$$

where $i=0, 1, 2, \dots, n-1$, and

$$\alpha_i = \arctan(2^{-i}) \quad (4)$$

the micro-rotation direction σ_i for the rotation mode ($z_n \rightarrow 0$) is $\sigma_i = \text{sign}(z_i)$, and for the vectoring mode ($y_n \rightarrow 0$) is $\sigma_i = -\text{sign}(x_i) \cdot \text{sign}(y_i)$.

Table 1 shows the elementary functions that can be evaluated by the CORDIC algorithm. For the i -th micro-rotation, a scaling factor becomes $k_i = \sqrt{1 + \sigma_i^2 2^{-2i}}$. After n micro-rotations, the product of all the scale factors is

$$K_1 = \prod_{i=0}^{n-1} k_i = \prod_{i=0}^{n-1} \sqrt{1 + \sigma_i^2 2^{-2i}} = \prod_{i=0}^{n-1} \sqrt{1 + 2^{-2i}} \quad (5)$$

where the rotation directions are defined by $\sigma_i = \{-1, +1\}$.

In a circular coordinate system, we let the iteration sequence of j be $0, 0, 0, 1, 2, \dots, n$, and

$$\theta_{\max} = \arctan(2^{-n}) + 2 \cdot \arctan(2^0) + \sum_{j=0}^n \arctan 2^{-j} \quad (6)$$

$$\cong 3.3141(189^\circ) > 180^\circ$$

According to eq. (6), the convergence range is expanded to $\pm 180^\circ$. Hence, the range of input can be unlimited [25] [26].

3. CORDIC-Based DCT Algorithm

1-D DCT Algorithm

The N -point 1-D DCT is defined as [1]

$$Y(m) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \sqrt{2} K_m \cos \left[\frac{(2n+1)m\pi}{2N} \right] \cdot x(n) \quad (7)$$

where $m = 0, \dots, N-1$,

$$K_m = \frac{1}{\sqrt{2}}, \text{ where } m = 0,$$

$$1, \text{ where } m > 0.$$

2-D DCT Algorithm

The $M \times N$ -point 2-D DCT is defined as

$$Z(u, v) = \frac{2 \cdot c(u)c(v)}{\sqrt{M \cdot N}}. \quad (8)$$

$$\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x(m, n) \cdot \cos\left[\frac{(2m+1)u\pi}{2M}\right] \cdot \cos\left[\frac{(2n+1)v\pi}{2N}\right]$$

where $u = 0, \dots, M-1, v = 0, \dots, N-1$,

$$c(k) = \frac{1}{\sqrt{2}}, \text{ where } k = 0,$$

$$1, \text{ where } k > 0.$$

According to eq. (8) we have

$$Z(u, v) = \frac{1}{\sqrt{M}} \sum_{m=0}^{M-1} \sqrt{2}c(u) \cdot \cos\left[\frac{(2m+1)u\pi}{2M}\right] \cdot \left\{ \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \sqrt{2}c(v) \cdot \cos\left[\frac{(2n+1)v\pi}{2N}\right] \cdot x(m, n) \right\} \quad (9)$$

where $u = 0, \dots, M-1, v = 0, \dots, N-1$,

$$c(k) = \frac{1}{\sqrt{2}}, \text{ where } k = 0,$$

$$1, \text{ where } k > 0.$$

Let $\mathbf{Z} = \mathbf{T}\mathbf{X}\mathbf{T}^t$ and $\mathbf{Y} = \mathbf{T}\mathbf{X}^t$, $\mathbf{Z}$ is represented as

$$\mathbf{Z} = \mathbf{T}\mathbf{Y}^t \quad (11)$$

Hence,

$$2\text{-D DCT}(\mathbf{X}) = 1\text{-D DCT}((1\text{-D DCT}(\mathbf{X}))^t) \quad (12)$$

Here, the transform matrix $\mathbf{T}$ is represented as

$$\mathbf{T} = \frac{1}{\sqrt{8}} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ a & c & d & f & -f & -d & -c & -a \\ b & e & -e & -b & -b & -e & e & b \\ c & -f & -a & -d & d & a & f & -c \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ d & -a & f & c & -c & -f & a & -d \\ e & -b & b & -e & -e & b & -b & e \\ f & -d & c & -a & a & -c & d & -f \end{bmatrix} \quad (13)$$

where

$$a = \sqrt{2} \cos\left(\frac{\pi}{16}\right), b = \sqrt{2} \cos\left(\frac{\pi}{8}\right), c = \sqrt{2} \cos\left(\frac{3\pi}{16}\right),$$

$$d = \sqrt{2} \cos\left(\frac{5\pi}{16}\right), e = \sqrt{2} \cos\left(\frac{3\pi}{8}\right), f = \sqrt{2} \cos\left(\frac{7\pi}{16}\right), \text{ and}$$

$$N = 8.$$

1-D DCT Fast Algorithm

The eq. (13) can be decomposed into eqs. (14) and (15), then 8-point 1-D DCT can be represented as [27]

$$\begin{bmatrix} Y(0) \\ Y(2) \\ Y(4) \\ Y(6) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ b & e & -e & -b \\ 1 & -1 & -1 & 1 \\ e & -b & b & -e \end{bmatrix} \begin{bmatrix} x(0) + x(7) \\ x(1) + x(6) \\ x(2) + x(5) \\ x(3) + x(4) \end{bmatrix} \quad (14)$$

$$\begin{bmatrix} Y(1) \\ Y(3) \\ Y(5) \\ Y(7) \end{bmatrix} = \begin{bmatrix} a & -c & d & -f \\ c & f & -a & d \\ d & a & f & -c \\ f & d & c & a \end{bmatrix} \begin{bmatrix} x(0) - x(7) \\ -x(1) + x(6) \\ x(2) - x(5) \\ -x(3) + x(4) \end{bmatrix} \quad (15)$$

Using eqs. (14) and (15), we can determine the data flow of 1-D DCT, which is shown in Figure 1. In Figure 1, we find the CORDIC(2) and CORDIC(5) have the same structure and rotation

angle of $\frac{\pi}{16}$ and the CORDIC(3) and CORDIC(4) have the same

structure and rotation angle of $\frac{5\pi}{16}$.

The CORDIC-Based Processor for 2-D DCT Architecture

The multiplication is the basic operator of the DCT architecture. The CORDIC-based processor performing the rotation mode in a circular coordinate system replaces the multiplier in the proposed DCT architecture. The arithmetic unit (AU) performed by the double-rotation CORDIC-based processor for DCT architecture [28]. Hence, compared with the conventional CORDIC-based arithmetic unit, the performance of the double-rotation CORDIC-based arithmetic unit is improved the latency more than 30% [28]. The hardware of arithmetic unit can be saved, so that the low-power consumption can be achieved. The overall relative error is less than 10^{-3} , when the bit-number of registers is defined by 16-bit, the number of micro-iterations of the double-rotation CORDIC processor is determined to be 8 [26].

4. The Proposed 2-D DCT Architecture

We use eq. (11), and propose the parallel-pipelined architecture for 8×8 2-D DCT processor which is shown in Figure 2. This architecture possesses two SRAM banks, two 8-point 1-D DCT processors, three multiplexer, and one 7-bit counter. 8-point 1-D DCT input-processor (P1) writes results to two SRAM banks alternately. 8-point 1-D DCT output-processor (P2) reads data from two SRAM banks alternately, and output the results.

Implementation of the Proposed 1-D DCT Processor

At first, the 8-point 1-D DCT processor would be introduced; the 1-D DCT processor has five CORDIC processors, which are accomplished by the double-rotation CORDIC arithmetic [27]. The proposed 1-D 8-point DCT architecture is shown in Figure 3. In this architecture, the DCT transformation matrix has a column-symmetry property, so that the shuffle structure is simplified and it requires no multipliers. The hardware is saved and the low power consumption is achieved.

Implementation of the Proposed 2-D DCT Processor

The latency of this processor is 64 clocks, the hardware complexity is $O(N \cdot \log_2 N)$ and the throughput is 8 outputs per cycle. The architecture without multiplier achieves small area and low-power. The propose architecture provides high throughput, and this work utilizes a parallel-pipelined architecture.

5. Conclusions

Based on the symmetry property considerations and fast DCT algorithm, a 1-D DCT computation algorithm has been proposed and implemented into VLSI parallel-pipelined architecture. In this 1-D DCT processor, the double-rotation CORDIC arithmetic were used for the implementation of the DCT arithmetic unit so as to perform multiply function. Further, a 2-D DCT processor has been proposed in accordance with the 2-D DCT computation decomposed into two successive 1-D DCT kernels. The architecture has simple and regular structure, are of trade-off designs, hence they are more scalable, flexible and efficient, and therefore very suitable for VLSI implementation. The comparison between the proposed

2-D DCT architecture and some existing 2-D DCT architectures is shown in Table 1.

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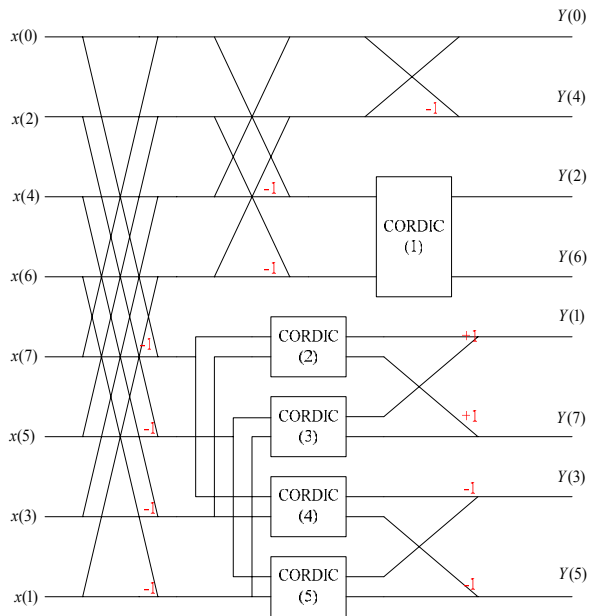


Fig. 1. 1-D 8-point DCT data flow

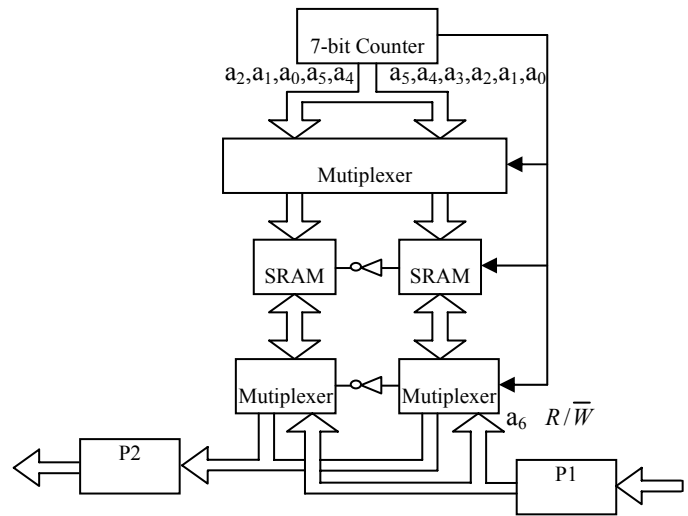


Fig. 2. The proposed 8x8 DCT processor

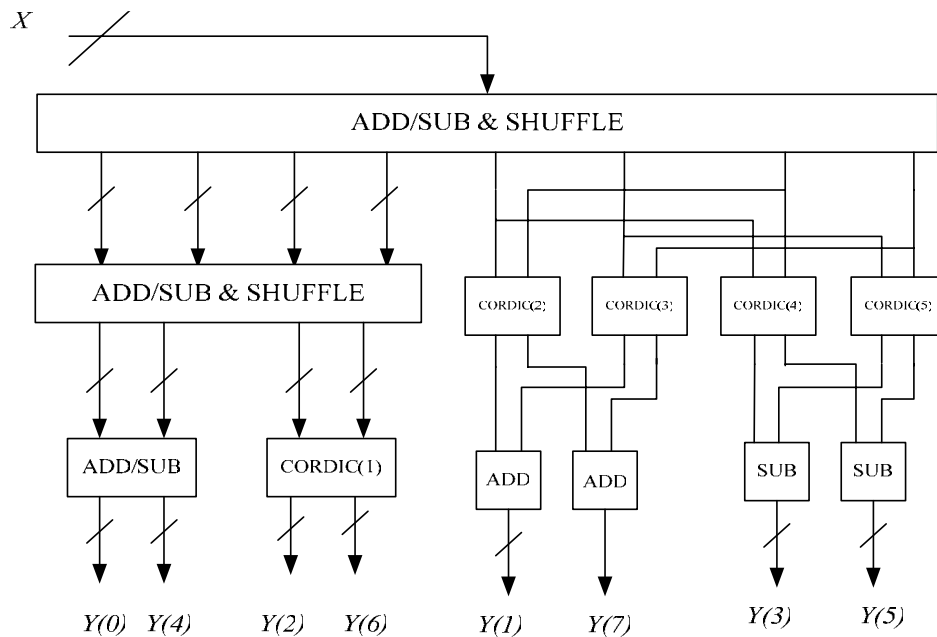


Fig. 3. The proposed 1-D 8-point DCT architecture (P1 and P2)

Table 1 Comparison between the proposed architecture and some existing architectures

8x8 2-D DCT	Lee [9]	Chang [10]	Hsiao [11]	Hsiao [12]	This Work
Real-multipliers	28	64	-	-	-
Real-adders	134	88	-	10	36
Complex-multipliers	-	-	3	3	-
Complex-adders	-	-	9	-	-
Delay elements (Words)	256	114	-	171	-
Memory (Words)	~384	~200	~370	-	134
Hardware complexity (AUs)	$O(N \log N)$	$O(N^2)$	$O(\log N)$	$O(\log N)$	$O(N \cdot \log_2 N)$
Throughput (outputs/cycle)	16	8	2	2	8
Pipelability	no	yes	yes	yes	yes